

Executive Business Report

Sales Forecasting and Demand Analytics Project

Executive Summary

This project analyzed historical sales data to understand business performance and predict future demand. The analysis identified sales trends, product demand patterns, and unusual sales activities. A forecasting system was developed to estimate future sales and support better inventory and supply chain decisions. The results help reduce stock shortages, avoid excess inventory, and improve business planning.

Key Findings from Sales Analysis and Forecasting

Historical sales analysis showed different demand patterns across time periods, product categories, and regions. Some products showed consistent demand while others had irregular sales behaviour.

Multiple forecasting methods were compared. The final selected forecasting model provided better prediction performance and was used to estimate future sales.

Three-Month Sales Forecast

Month 1 Forecast: 86,530

Expected Range: 75,000 - 98,000

Month 2 Forecast: 87,955

Expected Range: 76,000 - 100,000

Month 3 Forecast: 88,256

Expected Range: 77,000 - 101,000

Business Interpretation:

Sales are expected to remain stable with a gradual increase. The supply chain team should maintain suitable inventory levels and monitor actual sales against forecast values.

Top Three Sales Anomalies and Possible Causes

Anomaly 1:

An unusual sales increase was detected. Possible causes include large customer orders, seasonal demand, or promotional activities.

Anomaly 2:

An unexpected sales decrease was detected. Possible causes include lower customer demand, product availability issues, or market changes.

Anomaly 3:

Irregular product sales behaviour was identified. Possible causes include special orders, regional demand changes, or temporary market conditions.

Product Demand Segmentation Findings

High Demand Products:

These products have strong and regular demand.

Recommended stocking strategy: Maintain higher inventory levels and prioritize availability.

Medium Demand Products:

These products have stable but moderate demand.

Recommended stocking strategy: Maintain balanced inventory and review demand regularly.

Low Demand Products:

These products have lower and less predictable demand.

Recommended stocking strategy: Keep limited stock and avoid over-ordering.

Business Recommendations

1. Improve Inventory Planning:

Use sales forecasts to adjust inventory levels before demand changes. This can reduce stock shortages and unnecessary inventory costs.

2. Focus on High Performing Products:

Prioritize purchasing and warehouse space for products with stronger demand patterns.

3. Monitor Unusual Sales Events:

Investigate sudden sales changes to respond quickly and improve business planning.

System Limitation and Risk

The forecasting system depends on historical sales information. Unexpected events such as economic changes, supply chain disruptions, competitors, or sudden customer behaviour changes can reduce prediction accuracy.

Conclusion

This project provides a sales analytics solution that supports better forecasting, inventory planning, and data-driven business decisions.